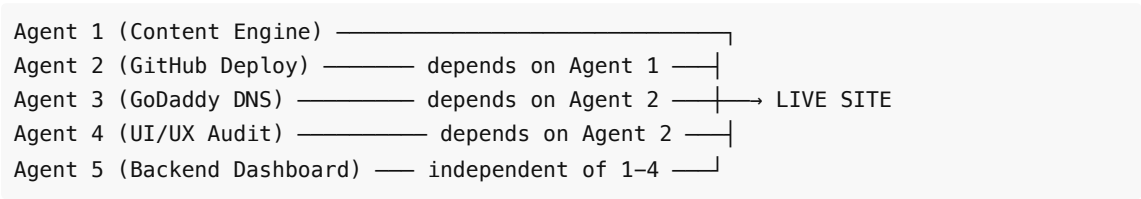


AGENTS.md — Business Simplifier Multi-Agent Workstream Registry

Last updated: 2026-05-25

This file documents the five specialised workstreams (agents) for the Business Simplifier project, their scope, I/O contracts, dependencies, and ordering.

Dependency Graph



Run Agent 1 → Agent 2 → Agents 3 & 4 in parallel → Agent 5 independently.

Agent 1: Content Engine

Scope: Generate and publish all 30 SEO-optimised blog posts to the Astro website.

Inputs:

- STRATEGY.md Section 6 (30-day content calendar)
- src/data/blogPosts.js (existing 9 posts as style reference)
- marketing.md (brand voice, keyword targets, competitor context)

Outputs:

- Updated src/data/blogPosts.js with 30 total blog posts
- Each post: { slug, title, excerpt, content, category, author, publishedDate, readTime, tags }
- Content requirements per post:
 - Minimum 400 words
 - At least 2 SA-specific statistics with source attribution
 - 1 CTA embedded (consultation booking)
 - Primary keyword in title, H2s, and first paragraph
 - No AI-slop: specific numbers, named businesses/regions, real ROI data

Conventions:

- Author: 'Business Simplifier Team'
- Date format: 'YYYY-MM-DD'
- Slugs: kebab-case, keyword-rich
- Tags: 3–5 per post, lowercase with spaces

Error Handling:

- If a stat cannot be verified, use hedging language ("research suggests", "industry estimates")
- Never fabricate specific company names as case study subjects without clear attribution

Testing:

- `npm run build` must complete with 0 errors
 - All 30 slugs must be unique
 - No undefined `category` values (must match `blogCategories` IDs)
-

Agent 2: GitHub Deploy

Scope: Initialise git repository, create GitHub Actions CI/CD pipeline, push to GitHub, enable GitHub Pages.

Inputs:

- Built Astro project at `AI Consultation/`
- GitHub username (provided by user)
- `STRATEGY.md` Section 7 (deploy workflow YAML)

Outputs:

- `.github/workflows/deploy.yml` created
- `public/CNAME` file with custom domain
- `git init` + first commit + remote pushed
- GitHub Pages enabled with GitHub Actions source

Shared Interfaces:

- Output: Live URL format `https://USERNAME.github.io/REPO_NAME` (pre-custom domain)
- Output: `CNAME` file content (used by Agent 3)

Conventions:

- Branch: `main` (not `master`)
- Commit message: `"feat: initial website deploy – Business Simplifier AI"`
- `.gitignore` must exclude: `node_modules/`, `dist/`, `.DS_Store`, `.env`

Error Handling:

- If Actions build fails: check `astro.config.mjs` site URL matches domain
- If Pages not activating: ensure repo is public

Dependencies: Agent 1 must be complete (all posts in `blogPosts.js`) before first deploy

Agent 3: GoDaddy DNS Configuration

Scope: Configure GoDaddy domain DNS records to point to GitHub Pages.

Inputs:

- GoDaddy account credentials (user provides)
- GitHub Pages IP addresses (from `STRATEGY.md` Section 7):
 - `185.199.108.153`
 - `185.199.109.153`
 - `185.199.110.153`
 - `185.199.111.153`
- Custom domain name (e.g., `businesssimplifier.co.za`)

Outputs:

- 4 x A records for root domain (@)
- 1 x CNAME record for `www` subdomain
- GitHub Pages HTTPS enforced
- SSL certificate issued automatically

Verification:

- DNS propagation check: `dig businesssimplifier.co.za A` → returns 4 GitHub IPs
- Online tool: dnschecker.org
- Expected propagation time: 15 min–48h

Dependencies: Agent 2 must have GitHub Pages live with custom domain set

Agent 4: UI/UX Audit & Implementation

Scope: Measure the live site against industry standards and implement priority improvements.

Inputs:

- Live site URL (post Agent 2 & 3 completion)
- `STRATEGY.md` Section 8 (audit checklist)
- Astro/Tailwind source files in `src/`

Outputs:

- PageSpeed Insights score ≥ 90 (mobile + desktop)
- All CWV metrics in green
- New components implemented:
 - `src/components/TestimonialsSection.astro`
 - `src/components/TrustBar.astro` (social proof numbers)
 - `src/components/StickyMobileCTA.astro`
 - `src/components/ExitIntentPopup.astro` (Alpine.js)
- Accessibility fixes applied across all components
- Internal linking added to blog posts

Priority Order (highest ROI first):

1. Sticky mobile CTA → estimated +23% mobile conversions
2. Testimonials section → builds trust, reduces bounce
3. Exit-intent popup → captures 5-8% abandoning visitors
4. CWV fixes → Google ranking factor
5. Accessibility pass → WCAG 2.1 AA compliance

Testing:

- PageSpeed Insights: <https://pagespeed.web.dev>
- WCAG: axe DevTools browser extension
- Mobile: Chrome DevTools → iPhone 14 viewport

Dependencies: Agent 2 complete (requires live URL for testing)

Agent 5: Backend Dashboard — AI Source Recommender

Scope: Design and build the AI-powered dashboard where sources are uploaded and AI implementation recommendations are generated.

Inputs:

- STRATEGY.md Section 9 (full architecture plan)
- Existing server.py and api/ directory (reference patterns)
- Claude API access (Anthropic SDK)

Outputs:

- dashboard/ directory with full project scaffold:
 - dashboard/backend/ — FastAPI application
 - dashboard/frontend/ — React + Vite application
- Working MVP features:
 - File upload (PDF, DOCX, XLSX, TXT, CSV)
 - Document parsing (PyMuPDF, python-docx, openpyxl)
 - Claude API integration with recommendation prompt
 - JSON response rendered as UI cards
 - Export to PDF
- Deployment config for Render.com (free tier)

I/O Contract (API):

```
POST /api/upload
Input: multipart/form-data { file: File }
Output: { document_id: str, filename: str, status: "uploaded" }

POST /api/analyse/{document_id}
Input: { document_id: str }
Output: {
  document_id: str,
  summary: str,
  processes: [
    {
      name: str,
      description: str,
      aiSolution: str,
      toolType: str,
      timeSavedHours: int,
      costEstimateZAR: int,
      difficulty: int (1-5),
      priority: int (1-10)
    }
  ]
}

GET /api/documents
Output: [{ document_id, filename, uploaded_at, analysed: bool }]
```

Shared Conventions (with existing codebase):

- Python: follow patterns in api/routes.py and api/services.py

- Error handling: HTTP exceptions with descriptive messages
- Logging: Python `logging` module, INFO level
- No hardcoded API keys — use environment variables (`ANTHROPIC_API_KEY`)

Dependencies:

- Independent of Agents 1–4
 - Requires `ANTHROPIC_API_KEY` environment variable
 - Node.js 22+ for frontend, Python 3.11+ for backend
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Shared Conventions (All Agents)

Naming

- Files: kebab-case for Astro/JS, snake_case for Python
- Components: PascalCase
- Variables: camelCase (JS), snake_case (Python)

Environment Variables

- Never commit `.env` files
- Always provide `.env.example` with key names and descriptions
- Document all required vars in README

Testing Mandate

- Every agent must run a build/lint check before marking complete
- Agent 1: `npm run build` with 0 errors
- Agent 2: GitHub Actions green checkmark
- Agent 3: DNS propagation confirmed at dnschecker.org
- Agent 4: PageSpeed score ≥ 90
- Agent 5: `uvicorn main:app` starts without errors

Error Escalation

- If blocked by missing credentials (GitHub, GoDaddy, Anthropic API key): document the blocker and provide exact instructions for the user to resolve it
- Never silently skip a step — note `⚠️ BLOCKED:` with clear resolution path